

Cryptography - 7_11

Today, we'll take a look at block ciphers.

As we know, PRFs are enough to do all the crypto in the MINICRYPT world, meaning SKE and MACs.

But we saw, sometimes, we need PRPs instead. Given k we can invert $F(k, x)$ efficiently.

In practice, PRPs are called block ciphers.

There are two main block ciphers:

- DES
- AES

Remark: Neither of them is provably secure based on standard assumptions.

However, their design is inspired by standard result in provable security.

For DES: It's the Feistel Network. It's a way to turn a PRF into a PRP.

Let $F : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a function. Consider the following permutation over $\{0, 1\}^{2n}$:

$$\Psi_F(x, y) = (y, x \oplus F(y)) = (x', y') \in \{0, 1\}^{2n}.$$

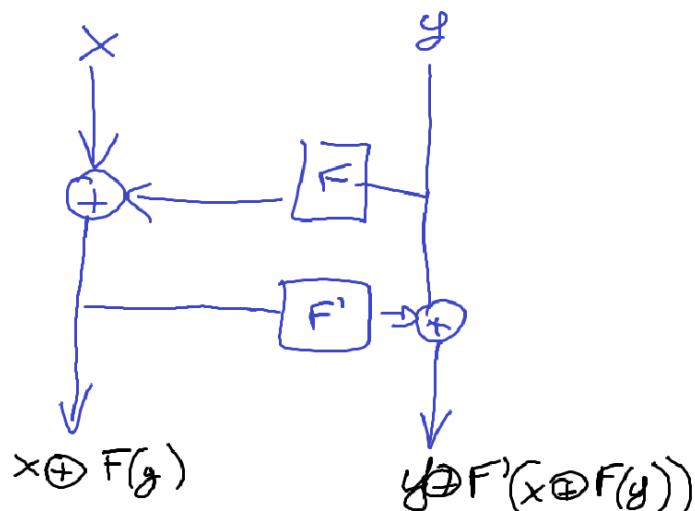
So I split the input in two halves. This can be inverted, in fact

$$\Psi'_F(x', y') = (F(x') \oplus y', x').$$

Is this pseudorandom? No, as it's distinguishable from a random string. If we replace F with R , then we can easily distinguish. So, no because the first $\frac{1}{2}$ of outputs is same as second $\frac{1}{2}$ of input.

The idea of Feistel is repeating this process multiple times.

For instance, let's do it two times:



So,

$$\Psi_{F,F'}(x, y) = \Psi_{F'}(\Psi_F(x, y))$$

In reality, F, F' are just $F(k_1, \cdot)$ and $F(k_2, \cdot)$ for independent keys k_1, k_2 .

Are two rounds enough? No, but almost, in fact there is a very specific kind of query that allows me to distinguish.

Note: $\Psi_{F,F'}(x, y) \oplus \Psi_{F,F'}(x', y)$ (so two queries with different x but same y , xored).

If I do this, we obtain $x \oplus x'$ for the first half, which is a problem. Then we can easily distinguish if we use R instead of F .

$\implies$ This implies the existence of a distinguisher against pseudorandomness.

The distinguisher will just give two queries, and check if $x \oplus x'$, if it is it outputs 1, else 0, with a probability of succeeding that is not negligible.

If I assume that all the queries are such that y are all different, then it's secure.

✓ Theorem

3 rounds of Feistel with a PRF (independent keys) are a PRP.

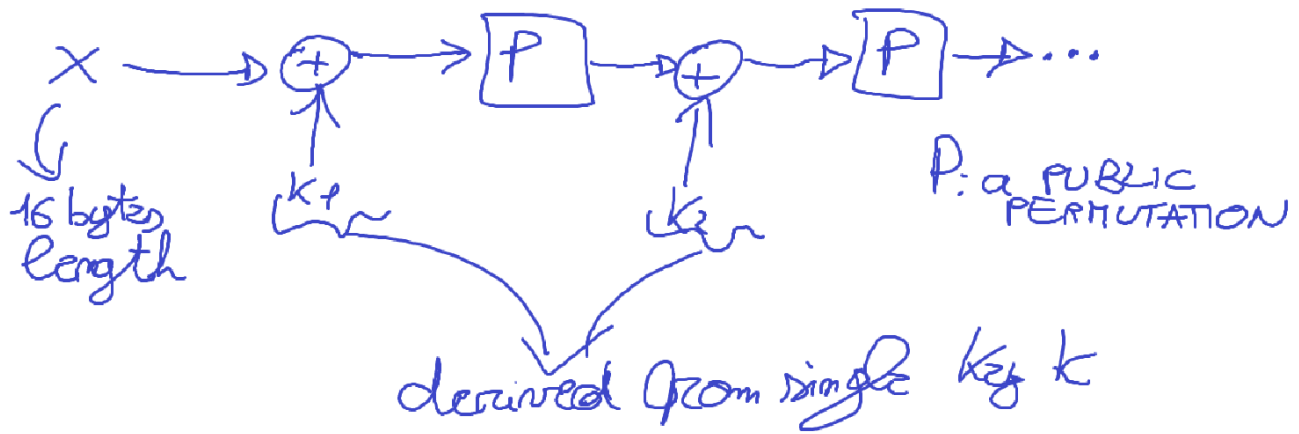
Intuition: Two rounds are enough if y 's are distinct. The first round makes the y 's distinct, except with small probability.

DES: Uses this construction with heuristic F , for 18 rounds and pseudorandom keys $k_1, \dots, k_{18}$ derived all from the same key k .

The main problem with DES is that the key is kind of short, this makes the brute-force approach possible. An Easy Fix: Double encryption, which however does not work, having the same brute force complexity, because of meet-in-the-middle attacks.

A solution: 3DES. It's not very efficient, for this reason a new standard was made: AES.

The provable security of AES is even worse: we know very little. This is an instantiation of substitution permutation networks (SPNs).



🔗 Lemma

For every unbounded attacker making $q \in \text{poly}(\lambda)$ queries the following are indistinguishable:

1. $S : \Psi_{F, F'}(\cdot, \cdot)$ for $F, F' \leftarrow \mathcal{R}(\lambda, n, n)$
2. $R : R \leftarrow \mathcal{R}(\lambda, 2n, 2n)$ so long as the queries $(x_1, y_1), \dots, (x_q, y_q)$ are YNIQUE (written like this yes, so with unique y's) ($y_i \neq y_j$ for $i \neq j$)

📄 Proof (Thm)

We consider some hybrids:

- $T : (x, y) \mapsto \Psi_{F_{k_3}}(\Psi_{F_{k_2}}(\Psi_{F_{k_1}}(x, y)))$, with $k_1, k_2, k_3 \leftarrow \mathcal{K}$.
- $S : (x, y) \mapsto \Psi_{F, F'}(\Psi_{F''}(x, y))$, with $F, F', F'' \leftarrow \mathcal{R}(\lambda, n, n)$
- $R : (x, y) \mapsto R(x, y)$, with $R \leftarrow \mathcal{R}(\lambda, 2n, 2n)$, which is a random function.
- $P : (x, y) \mapsto P(x, y)$, with $P \leftarrow \mathcal{P}(\lambda, 2n, 2n)$, which is a random permutation.

Starting with T , we need to prove it's indistinguishable from $\mathcal{P}$, and we do it through some passages.

$T(\lambda) \approx_c S(\lambda)$ is a simple reduction to PRF security. We have to create some hybrid arguments, as we have to switch the three PRFs one at a time.

$R(\lambda) \approx_c T(\lambda)$:

$$\Pr[\exists i, j \text{ s.t. } i \neq j : R(x_i, x_j) = R(x_i, y_i)] \underset{\text{union bound}}{\leq} \binom{q}{2} \cdot 2^{-2n} = \text{negl}(n)$$

Now, I need to prove $S(\lambda) \approx_c R(\lambda)$. Here, we use the lemma. How? We prove that for every $(x_1, y_1), \dots, (x_q, y_q)$ the probability that the output $\Psi_{F''}(x_i, y_i)$ are not unique is negligible. Then, we can invoke the lemma.

Let $(x_i, y_i), (x_j, y_j)$ be the inputs such that $(x_i, y_i) \neq (x_j, y_j)$.

Assume first that $y_i = y_j$, so $x_i \neq x_j$ (to respect the property stated before). Then $y''_i = x_i \oplus F''(y_i)$ (with y''_i the output), which is $\neq x_j \oplus F''(y_i) = x_j \oplus F''(y_j) = y''_j$.

So long as y_i is equal to y_j , the outputs are always unique

Now, assume $y_i \neq y_j$, then $x_i \oplus x_j = F''(y_i) \oplus F''(y_j)$.

$$\implies \Pr[F''(y_i) \oplus F''(y_j) = x_i \oplus x_j] \leq 2^{-n}$$

$$\implies \Pr[\neg \text{unique}] \leq \binom{q}{2} \cdot 2^{-n} = \text{negl}(n)$$

This implies that $T(\lambda) \approx_c S(\lambda) \approx_c R(\lambda) \approx_c P(\lambda)$

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